

Multiple Linear Regression

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From SLR to MLR

Recap: simple linear regression

The model with one regressor:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

OLS estimators, written through the sample covariance, the variances and the correlation coefficient r :

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\widehat{\text{Cov}}(x, y)}{s_x^2} = r \frac{s_y}{s_x}, \quad (1)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - 2}. \quad (2)$$

Question: is one regressor enough?

Delivery time of a soft-drink route driver depends on the number of cases loaded (x_1) and the walking distance (x_2).

Why not simply fit two separate SLRs, y on x_1 and y on x_2 ?

- The two regressors are typically correlated (more cases usually means longer distance).
- Each SLR slope then absorbs part of the effect of the omitted regressor, so it is not the effect of that regressor alone.
- Separate fits cannot answer “what is the effect of one more case for a fixed distance?”.
- The residual variance is also larger because the omitted regressor’s effect is left in the error.

Example: delivery time data

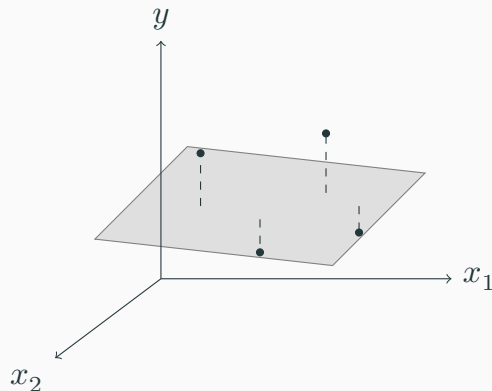
Montgomery, Peck & Vining, Example 3.1 ($n = 25$).

- y : delivery time (minutes)
- x_1 : number of cases stocked
- x_2 : distance walked by the driver (feet)

Obs.	Time y	Cases x_1	Distance x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
$\vdots$	$\vdots$	$\vdots$	$\vdots$
25	10.75	4	150

From a line to a plane

- SLR: a line in the (x, y) plane.
- Two regressors: a plane in (x_1, x_2, y) space.
- k regressors: a hyperplane in $(k + 1)$ dimensions.
- Residuals: vertical distances to the fitted surface.



The MLR model

The MLR equation

With k regressors and n observations:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \dots, n. \quad (3)$$

- x_{ij} : value of regressor j for observation i .
- $p = k + 1$ unknown regression coefficients $\beta_0, \dots, \beta_k$ (including the intercept).
- ε_i : random error of observation i .

Meaning of the coefficients

$$\frac{\partial E[y \mid x_1, \dots, x_k]}{\partial x_j} = \beta_j, \quad j = 1, \dots, k.$$

- β_j is the change in the mean response per unit increase in x_j *when all other regressors are held fixed*.
- For this reason $\beta_1, \dots, \beta_k$ are called partial regression coefficients.
- β_0 is the mean response when all regressors are zero (meaningful only if that point is within the range of the data).

Question: interpreting the coefficients

In the delivery time model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, interpret β_1 , β_2 and β_0 in words, with units.

From n equations to one matrix equation

Writing (3) out for every observation:

$$y_1 = \beta_0 + \beta_1 x_{11} + \cdots + \beta_k x_{1k} + \varepsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \cdots + \beta_k x_{2k} + \varepsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_{n1} + \cdots + \beta_k x_{nk} + \varepsilon_n$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into a matrix.

The MLR model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (4)$$

$$\underbrace{\mathbf{y}}_{n \times 1}, \quad \underbrace{\mathbf{X}}_{n \times p}, \quad \underbrace{\boldsymbol{\beta}}_{p \times 1}, \quad \underbrace{\boldsymbol{\varepsilon}}_{n \times 1}.$$

The vectors $\mathbf{y}$, $\boldsymbol{\beta}$, $\boldsymbol{\varepsilon}$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The design matrix X

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}$$

- Row i is observation i ; column j is regressor j .
- The first column of ones corresponds to the intercept β_0 .

Model assumptions

$$E[\boldsymbol{\varepsilon}] = \mathbf{0}, \quad (5)$$

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n. \quad (6)$$

- $\mathbf{X}$ is fixed (non-random) and has full column rank, i.e. $\text{rank}(\mathbf{X}) = p \leq n$.
- No distributional assumption on $\boldsymbol{\varepsilon}$ is needed yet.

What does $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ say?

$$\text{Var}(\boldsymbol{\varepsilon}) = E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \begin{pmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{pmatrix}$$

- Diagonal entries: every error has the same variance σ^2 (constant variance).
- Off-diagonal entries: errors of different observations are uncorrelated.

“Linear” means linear in the parameters

Each of these is an MLR model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon,$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon.$$

This one is not:

$$y = \beta_0 e^{\beta_1 x} + \varepsilon.$$

Polynomial and interaction terms are handled later in the course.

OLS estimation in matrix form

The least-squares criterion

Choose β to minimise the sum of squared errors

$$S(\beta) = \sum_{i=1}^n \varepsilon_i^2 = \boldsymbol{\varepsilon}' \boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta). \quad (7)$$

Expanding $S(\beta)$

$$\begin{aligned} S(\beta) &= \mathbf{y}'\mathbf{y} - \beta'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta'\mathbf{X}'\mathbf{X}\beta \\ &= \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta. \end{aligned} \tag{8}$$

The second line combines the two middle terms.

Question: combining the middle terms

Why is $\mathbf{y}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{X}'\mathbf{y}$?

Two rules for differentiating with respect to a vector

For a constant vector $\mathbf{a}$ and a constant symmetric matrix $\mathbf{A}$:

$$\frac{\partial \mathbf{a}'\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = \mathbf{a}, \quad \frac{\partial \boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = 2\mathbf{A}\boldsymbol{\beta}.$$

Apply with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric) to (8).

The normal equations

$$\begin{aligned}\frac{\partial S}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\beta, \\ \left. \frac{\partial S}{\partial \beta} \right|_{\hat{\beta}} &= \mathbf{0}, \\ \mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{y}.\end{aligned}\tag{9}$$

This is a system of p linear equations in the p unknowns $\hat{\beta}_0, \dots, \hat{\beta}_k$.

The OLS estimator

If $\mathbf{X}'\mathbf{X}$ is invertible, the normal equations (9) have the unique solution

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \quad (10)$$

When is $\mathbf{X}'\mathbf{X}$ invertible?

- $\mathbf{X}'\mathbf{X}$ ($p \times p$) is invertible if and only if $\mathbf{X}$ has full column rank.
- That requires $n \geq p$ and no column of $\mathbf{X}$ being a linear combination of the other columns.
- If a regressor is (nearly) a linear combination of the others, $\mathbf{X}'\mathbf{X}$ is singular (nearly singular): this is the collinearity problem, treated in a later week.

Is $\hat{\beta}$ a minimum?

$$\frac{\partial^2 S}{\partial \beta \partial \beta'} = 2\mathbf{X}'\mathbf{X}.$$

- For any $\mathbf{a} \neq \mathbf{0}$: $\mathbf{a}'\mathbf{X}'\mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})'(\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|^2 > 0$, because full column rank gives $\mathbf{X}\mathbf{a} \neq \mathbf{0}$.
- So $\mathbf{X}'\mathbf{X}$ is positive definite and $\hat{\beta}$ is the unique minimiser of (7).

Check: the SLR case, ingredients

For SLR, $\mathbf{X} = [\mathbf{1} \ \mathbf{x}]$ and

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

Check: the SLR case, solution

Since $n \sum x_i^2 - (\sum x_i)^2 = nS_{xx}$,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}.$$

The second entry of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is

$$\hat{\beta}_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \frac{S_{xy}}{S_{xx}},$$

which is (1).

Fitted values, residuals and the hat matrix

Fitted values and residuals

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad (11)$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y},$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \quad (12)$$

$\mathbf{H}$ is the $n \times n$ hat matrix: it puts the hat on $\mathbf{y}$.

Properties of the hat matrix

$$\mathbf{H}' = \mathbf{H}, \quad (13)$$

$$\mathbf{H}^2 = \mathbf{H}, \quad (14)$$

$$\mathbf{H}\mathbf{X} = \mathbf{X}. \quad (15)$$

That is, $\mathbf{H}$ is symmetric and idempotent.

Question: idempotency of H

Prove (14) from the definition (11).

Properties of $\mathbf{I} - \mathbf{H}$

From (13), (14) and (15):

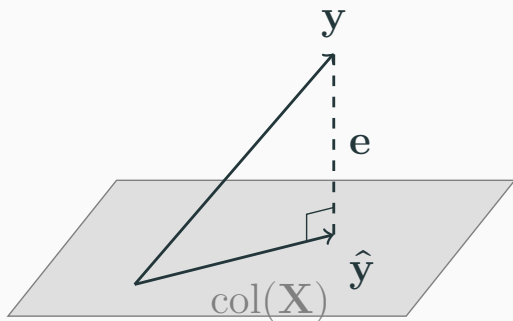
$$(\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}, \quad (16)$$

$$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}, \quad (17)$$

$$(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}. \quad (18)$$

Geometry of least squares

- The columns of $\mathbf{X}$ span a p -dimensional subspace $\text{col}(\mathbf{X})$ of $\mathbb{R}^n$.
- $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$.
- $\mathbf{e}$ is perpendicular to $\text{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}$ is the point of $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$, so $\hat{\boldsymbol{\beta}}$ minimises $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$.



Orthogonality of residuals and regressors

$$\mathbf{X}'\mathbf{e} = \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \quad (19)$$

by the normal equations (9). Consequences:

- First row of $\mathbf{X}'$ is a row of ones, so $\sum_i e_i = 0$.
- Row j gives $\sum_i x_{ij}e_i = 0$ for each regressor.
- $\hat{\mathbf{y}}'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = 0$.

Delivery time example: OLS fit

Computing $X'X$ and $X'y$

$$X'X = \begin{pmatrix} 25 & 219 & 10\,232 \\ 219 & 3\,055 & 133\,899 \\ 10\,232 & 133\,899 & 6\,725\,688 \end{pmatrix},$$

$$X'y = \begin{pmatrix} 559.60 \\ 7\,375.44 \\ 337\,071.69 \end{pmatrix}.$$

The inverse

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

The fitted equation

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{pmatrix} 2.3412 \\ 1.6159 \\ 0.01438 \end{pmatrix}$$

$$\hat{y} = 2.3412 + 1.6159 x_1 + 0.01438 x_2.$$

Interpreting the fit

- Each additional case adds about 1.62 minutes to the delivery time, for a fixed distance.
- Each additional foot adds about 0.0144 minutes (about 1.4 minutes per 100 ft), for a fixed number of cases.
- The SLR of y on x_1 alone gives a slope of 2.18, not 1.62: cases and distance are correlated ($r = 0.82$), and the SLR slope absorbs part of the distance effect.

Verifying in R

```
y <- delivery$time  
X <- cbind(1, delivery$cases, delivery$distance)
```

```
solve(t(X) %*% X, t(X) %*% y)
```

```
coef(lm(time ~ cases + distance,  
        data = delivery))
```

- `solve(A, b)` solves $\mathbf{Ax} = \mathbf{b}$ directly; it is numerically preferable to forming $\mathbf{A}^{-1}$ explicitly.

Properties of the OLS estimators

Linearity

$$\hat{\beta} = \mathbf{A}\mathbf{y}, \quad \mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \quad (20)$$

$\mathbf{A}$ is a $p \times n$ matrix of constants (given $\mathbf{X}$), so every $\hat{\beta}_j$ is a linear combination of $y_1, \dots, y_n$.

Decomposing $\hat{\beta}$

Substitute the model (4) into (10):

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon) \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon.\end{aligned}\tag{21}$$

The estimator is the true β plus a linear function of the errors.

Unbiasedness

Taking expectations in (21) with $\mathbf{X}$ fixed, and using (5):

$$\begin{aligned} E[\hat{\beta}] &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\varepsilon] \\ &= \beta. \end{aligned}$$

The covariance matrix of a random vector

$$\text{Var}(\hat{\beta}) = E[(\hat{\beta} - E[\hat{\beta}])(\hat{\beta} - E[\hat{\beta}])']$$

is a $p \times p$ symmetric matrix:

- diagonal entries: $\text{Var}(\hat{\beta}_j)$;
- off-diagonal entries: $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j)$.

For a constant matrix $\mathbf{B}$ and a random vector $\mathbf{z}$, directly from the definition,

$$\text{Var}(\mathbf{Bz}) = \mathbf{B} \text{Var}(\mathbf{z}) \mathbf{B}'. \quad (22)$$

Covariance matrix of $\hat{\beta}$

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \text{Var}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{Var}(\boldsymbol{\varepsilon})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.\end{aligned}\tag{23}$$

The steps use (21) (the constant β adds no variance), (22) and (6).

Reading the covariance matrix

Let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$, with entries C_{ij} , $i, j = 0, \dots, k$.

$$\begin{aligned}\text{Var}(\hat{\beta}_j) &= \sigma^2 C_{jj}, \\ \text{Cov}(\hat{\beta}_i, \hat{\beta}_j) &= \sigma^2 C_{ij}, \quad i \neq j.\end{aligned}\tag{24}$$

Check: the SLR case

Using the inverse computed for SLR, with $\sum x_i^2 = S_{xx} + n\bar{x}^2$:

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \frac{n}{nS_{xx}} = \frac{\sigma^2}{S_{xx}},$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \frac{\sum x_i^2}{nS_{xx}} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right),$$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \frac{\sum x_i}{nS_{xx}} = -\sigma^2 \frac{\bar{x}}{S_{xx}}.$$

These are the familiar SLR results.

Question: what makes $\text{Var}(\hat{\beta}_j)$ large?

Using (24), which features of the data and the design make the variance of $\hat{\beta}_j$ large?

The Gauss–Markov theorem

Gauss–Markov theorem

Under (5) and (6) (no normality needed), the OLS estimator $\hat{\beta}$ is the best linear unbiased estimator (BLUE) of β .

- Linear: a linear function of $\mathbf{y}$.
- Unbiased: $E[\hat{\beta}] = \beta$.
- Best: among all linear unbiased estimators, each $\hat{\beta}_j$, and more generally each $\mathbf{c}'\hat{\beta}$, has the smallest variance.

Distribution under normal errors

Distribution of $\hat{\beta}$

Add the assumption $\epsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. By (21), $\hat{\beta}$ is a linear function of a normal vector, hence normal:

$$\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}). \quad (25)$$

The mean and covariance matrix are the ones derived above. This result drives the inference procedures in the coming weeks.

Maximum likelihood under normal errors

The log-likelihood of (β, σ^2) is

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)' (\mathbf{y} - \mathbf{X}\beta). \quad (26)$$

For any fixed σ^2 , maximising over β is the same as minimising $S(\beta)$ in (7), so the ML estimator of β equals the OLS estimator.

Maximum likelihood estimator of σ^2

Setting $\partial \ell / \partial \sigma^2 = 0$ at $\beta = \hat{\beta}$ gives

$$\hat{\sigma}_{\text{ML}}^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})}{n} = \frac{\text{SSE}}{n}. \quad (27)$$

We show next that this estimator is biased.

Estimating the variance parameter σ^2

Residual sum of squares in matrix form

$$\begin{aligned}\text{SSE} &= \mathbf{e}'\mathbf{e} = \mathbf{y}'(\mathbf{I} - \mathbf{H})'(\mathbf{I} - \mathbf{H})\mathbf{y} \\ &= \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y}.\end{aligned}\tag{28}$$

The first line uses (12); the second uses (16) and (17).

Residuals as a function of the errors

$$\begin{aligned}\mathbf{e} &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon},\end{aligned}\tag{29}$$

$$\text{SSE} = \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}.\tag{30}$$

The first line uses (18); (30) follows as in (28), now with (29).

Expectation of a quadratic form

For a constant symmetric matrix $\mathbf{B}$:

$$\begin{aligned} E[\boldsymbol{\varepsilon}' \mathbf{B} \boldsymbol{\varepsilon}] &= E[\text{tr}(\boldsymbol{\varepsilon}' \mathbf{B} \boldsymbol{\varepsilon})] = E[\text{tr}(\mathbf{B} \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}')] \\ &= \text{tr}(\mathbf{B} E[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}']) = \sigma^2 \text{tr}(\mathbf{B}). \end{aligned} \tag{31}$$

- A scalar equals its trace.
- $\text{tr}(\mathbf{MN}) = \text{tr}(\mathbf{NM})$, and E commutes with tr .
- $E[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}'] = \sigma^2 \mathbf{I}_n$ by (5) and (6).

The trace of $\mathbf{I} - \mathbf{H}$

$$\begin{aligned}\text{tr}(\mathbf{H}) &= \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) \\ &= \text{tr}(\mathbf{I}_p) = p, \\ \text{tr}(\mathbf{I}_n - \mathbf{H}) &= n - p.\end{aligned}\tag{32}$$

$E[\text{SSE}]$ and an unbiased estimator of σ^2

Applying (31) with $\mathbf{B} = \mathbf{I} - \mathbf{H}$ to (30), and using (32):

$$E[\text{SSE}] = \sigma^2(n - p), \quad (33)$$

$$\hat{\sigma}^2 = \text{MSE} = \frac{\text{SSE}}{n - p}. \quad (34)$$

$E[\text{MSE}] = \sigma^2$. For $p = 2$ this is the SLR estimator (2).

Question: why divide by $n - p$?

The n residuals are $e_i = y_i - \hat{y}_i$. Looking at (33), why is the divisor of SSE $n - p$ and not n ? What is the bias of $\hat{\sigma}_{\text{ML}}^2$ in (27)?

Standard errors of the coefficients

Replace σ^2 by $\hat{\sigma}^2$ in (23) and (24):

$$\begin{aligned}\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) &= \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}, \\ \text{se}(\hat{\beta}_j) &= \sqrt{\hat{\sigma}^2 C_{jj}}.\end{aligned}\tag{35}$$

Delivery time data: $\hat{\sigma}^2$

- $SSE = 233.73$, with $n - p = 25 - 3 = 22$ degrees of freedom.
- $\hat{\sigma}^2 = MSE = 233.73/22 = 10.62$.
- $\hat{\sigma} = 3.26$ minutes: the typical deviation of a delivery time from the fitted plane.

Delivery time data: standard errors

Using (35):

Coefficient	Estimate	Standard error
β_0 (intercept)	2.3412	1.0967
β_1 (cases)	1.6159	0.1707
β_2 (distance)	0.01438	0.003613

Coefficient of determination and adjusted R^2

Sums of squares

$$\begin{aligned}\text{SST} &= \sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2, \\ \text{SSE} &= \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}, \\ \text{SSR} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - n\bar{y}^2, \\ \text{SST} &= \text{SSR} + \text{SSE}.\end{aligned}\tag{36}$$

The decomposition follows from the orthogonality (19).

The coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \quad (37)$$

- The proportion of the variability in y explained by the regressors;
 $0 \leq R^2 \leq 1$.
- Delivery time data: $R^2 = 0.9596$.

Question: adding a regressor

What happens to R^2 when a regressor is added to the model? Does it depend on whether the regressor is relevant?

Adding an irrelevant regressor

Add z , 25 independent $N(0, 1)$ draws unrelated to the delivery time.

Model	p	R^2	MSE
cases	2	0.9305	17.484
cases, distance	3	0.9596	10.624
cases, distance, z	4	0.9597	11.109

R^2 increases slightly with the useless regressor, while the MSE goes up.

Adjusted R^2

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n - p)}{\text{SST}/(n - 1)} = 1 - \frac{\text{MSE}}{\text{MST}}, \quad (38)$$

$$= 1 - \frac{n - 1}{n - p} (1 - R^2). \quad (39)$$

$\text{MST} = \text{SST}/(n - 1)$ is the sample variance of y .

Why adjusted R^2 penalises useless regressors

- MST does not depend on the model, so by (38) R^2_{adj} increases if and only if MSE decreases.
- Adding a regressor lowers SSE but also lowers the degrees of freedom $n - p$ in (34).
- The MSE decreases only if the drop in SSE outweighs the lost degree of freedom.

Adjusted R^2 in the example

Model	p	R^2	R^2_{adj}
cases	2	0.9305	0.9275
cases, distance	3	0.9596	0.9559
cases, distance, z	4	0.9597	0.9539

R^2_{adj} rises when the relevant regressor (distance) is added and falls when the irrelevant z is added.

Properties of adjusted R^2

From (39):

- For $p > 1$, $R_{\text{adj}}^2 < R^2$ unless $R^2 = 1$.
- It can decrease when a regressor is added.
- It can be negative: $R_{\text{adj}}^2 < 0$ whenever $R^2 < (p - 1)/(n - 1)$.

Caveats

- A high R^2 or R^2_{adj} does not mean the model is adequate: the assumptions must be checked with residual analysis.
- R^2_{adj} is a descriptive measure, not a hypothesis test.
- Formal comparison of models is done later with F-tests and information criteria.

Summary

Summary

- Model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $E[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ (4).
- OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ (10); $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto $\text{col}(\mathbf{X})$.
- $\hat{\boldsymbol{\beta}}$ is unbiased with $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (23), and is BLUE.
- $\hat{\sigma}^2 = \text{SSE}/(n - p)$ is unbiased for σ^2 (34).
- R^2 never decreases when regressors are added; R_{adj}^2 (38) penalises the loss of degrees of freedom.

Next lecture

- Inference on the regression coefficients: t -tests and confidence intervals.
- Test for significance of regression (F-test).
- Reading: Montgomery, Peck & Vining, Chapter 3.

Proof of the Gauss–Markov Theorem

Gauss–Markov: setup of the proof

Any linear estimator can be written

$$\tilde{\beta} = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$$

for some $p \times n$ matrix $\mathbf{D}$. Its expectation is $\beta + \mathbf{D}\mathbf{X}\beta$, so unbiasedness for every β requires

$$\mathbf{D}\mathbf{X} = \mathbf{0}. \tag{40}$$

Gauss–Markov: the variance comparison

$$\begin{aligned}\text{Var}(\tilde{\beta}) &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}] [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'] \\ &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{D}\mathbf{X})' \\ &\quad + \mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}'] \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}',\end{aligned}$$

where the last step uses (40).

Gauss–Markov: conclusion

- $\mathbf{a}'\mathbf{D}\mathbf{D}'\mathbf{a} = \|\mathbf{D}'\mathbf{a}\|^2 \geq 0$ for every $\mathbf{a}$, so $\mathbf{D}\mathbf{D}'$ is positive semidefinite.
- Hence $\text{Var}(\tilde{\boldsymbol{\beta}}) - \text{Var}(\hat{\boldsymbol{\beta}})$ is positive semidefinite, and $\text{Var}(\mathbf{c}'\tilde{\boldsymbol{\beta}}) \geq \text{Var}(\mathbf{c}'\hat{\boldsymbol{\beta}})$ for every $\mathbf{c}$.
- Equality holds only for $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$.

Questions?
